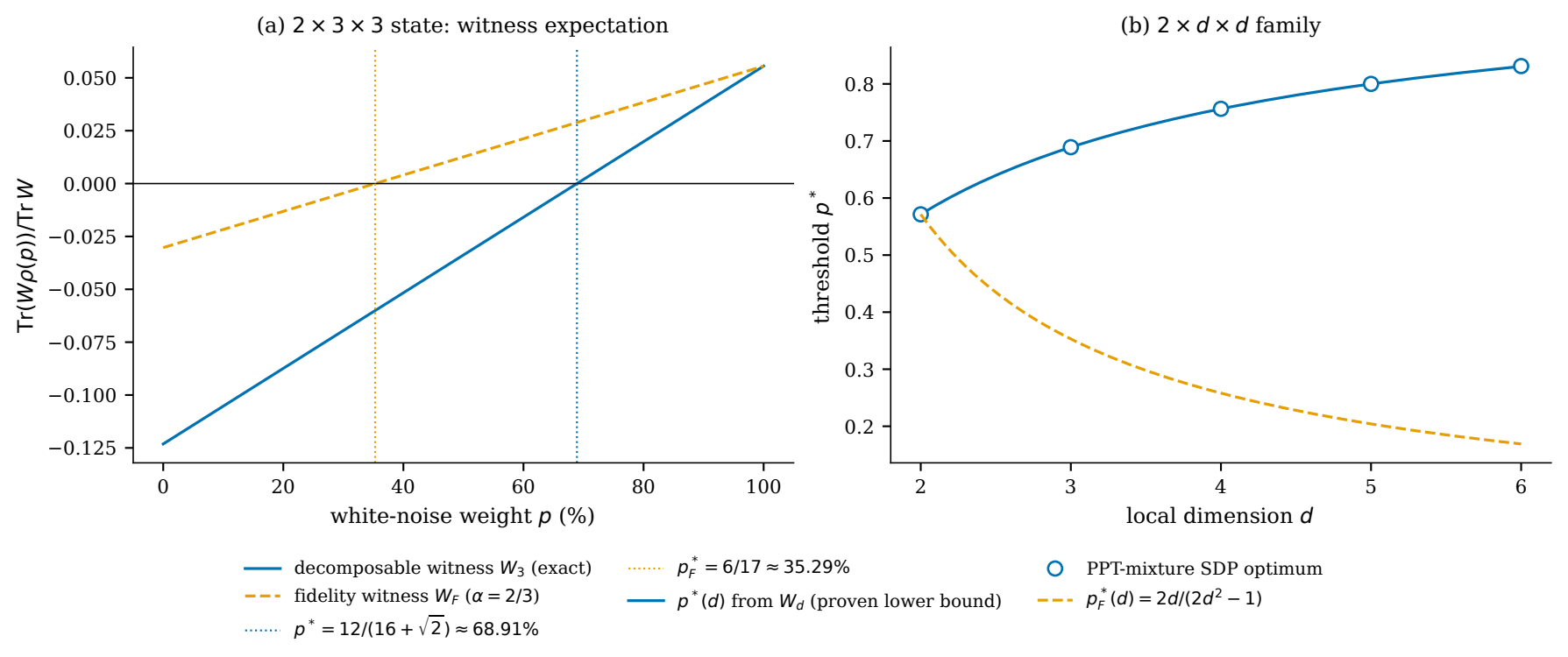

(a) 2 × 3 × 3 state: witness expectation
Tr(Wρ(p))/Tr W
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
−0.125
0
20
40
60
80
100
white-noise weight p (%)
(b) 2 × d × d family
threshold p*
0.8
0.7
0.6
0.5
0.4
0.3
0.2
2
3
4
5
6
local dimension d
decomposable witness W3 (exact)
fidelity witness WF (α = 2/3)
p* = 12/(16 + √2) ≈ 68.91%
p*F = 6/17 ≈ 35.29%
p*(d) from Wd (proven lower bound)
PPT-mixture SDP optimum
p*F(d) = 2d/(2d² − 1)